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Oefeningenreeks 4A nr. 3.12

$$\begin{aligned} & \int 4x\sqrt[3]{3x}dx \\ &= \int 4x \cdot 3^{\frac{1}{3}} \cdot x^{\frac{1}{3}}dx = 4 \cdot 3^{\frac{1}{3}} \int x^{1+\frac{1}{3}}dx = 4 \cdot 3^{\frac{1}{3}} \int x^{\frac{4}{3}}dx \\ &= 4 \cdot 3^{\frac{1}{3}} \cdot \frac{3}{7}x^{\frac{7}{3}} + C = \frac{12}{7}3^{\frac{1}{3}}x^{\frac{7}{3}} + C \\ &= \frac{12}{7}\sqrt[3]{3} \cdot x^2\sqrt[3]{x} + C = \frac{12}{7}x^2\sqrt[3]{3x} + C \end{aligned}$$

References